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Camera optical lens including six lenses of $-++-+-$ refractive powers

Abstract

The present disclosure relates to the technical field of optical lens and discloses a camera optical lens. The camera optical lens includes, from an object side to an image side: a first lens having a negative refractive power; a second lens having a positive refractive power; a third lens having a positive refractive power; a fourth lens having a negative refractive power; a fifth lens having a positive refractive power; and a sixth lens having a negative refractive power. The camera optical lens satisfies following conditions: $-8.00 \leq f1/f \leq -3.00$; $-6.00 \leq f34/f \leq -1.50$; and $R10/R9 \geq 1.50$. The camera optical lens has outstanding optical functions, while satisfying a desire of large aperture and long focal length.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to the field of optical lens, in particular, to a camera optical lens suitable for handheld devices, such as smart phones and digital cameras, and imaging devices, such as monitors, PC lenses or vehicle-mounted lenses.

BACKGROUND

(2) With the emergence of various smart devices in recent years, the demand for miniature camera optical lens is increasing day by day, and as the pixel size of the photosensitive devices become smaller, plus the current development trend of electronic products towards better functions and thinner and portable dimensions, miniature camera optical lens with good imaging quality therefore have become a mainstream in the market. In order to obtain better imaging quality, the lens generally adopts a multi-piece lens structure. Also, with the development of technology and the increase of the diverse demands of users, and as the pixel area of photosensitive devices is becoming smaller and smaller and the requirement of the system on the imaging quality is improving constantly, the six-piece lens structure gradually appear in lens designs. There is an urgent need for long-focal-length camera lenses with good optical characteristics, large aperture and fully corrected aberration.

SUMMARY

(3) To address the above issues, the present disclosure seeks to provide a camera optical lens that satisfies a design requirement of large aperture and long focal length while having outstanding optical functions.

(4) In order to address the above issues, embodiments of the present disclosure provide a camera optical lens including, from an object side to an image side: a first lens having a negative refractive power; a second lens having a positive refractive power; a third lens having a positive refractive power; a fourth lens having a negative refractive power; a fifth lens having a positive refractive power; and a sixth lens having a negative refractive power; wherein the camera optical lens satisfies following conditions: $-8.00 \leq f_1/f \leq -3.00$; $-6.00 \leq f_{34}/f \leq -1.50$; and $R_{10}/R_9 \geq 1.50$; where f denotes a focal length of the camera optical lens; f_1 denotes a focal length of the first lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; R_9 denotes a central curvature radius of an object-side surface of the fifth lens; and R_{10} denotes a central curvature radius of an image-side surface of the fifth lens.

(5) As an improvement, the camera optical lens further satisfies the following condition: $2.00 \leq R_{12}/R_{11} \leq 30.00$; where R_{11} denotes a central curvature radius of an object-side surface of the sixth lens; and R_{12} denotes a central curvature radius of an image-side surface of the sixth lens.

(6) As an improvement, the camera optical lens further satisfies the following condition: $5.00 \leq d_8/d_4 \leq 20.00$; where d_4 denotes an on-axis distance from an image-side surface of the second lens to an object-side surface of the third lens; and d_8 denotes an on-axis distance from an image-side surface of the fourth lens to an object-side surface of the fifth lens.

(7) As an improvement, an object-side surface of the first lens is convex in a paraxial region and an image-side surface of the first lens is concave in the paraxial region; and the camera optical lens further satisfies following conditions: $2.11 \leq (R_1 + R_2)/(R_1 - R_2) \leq 7.56$; and $0.04 \leq d_1/\text{TTL} \leq 0.22$; where R_1 denotes a central curvature radius of an object-side surface of the first lens; R_2 denotes a central curvature radius of an image-side surface of the first lens; d_1 denotes an on-axis thickness of the first lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(8) As an improvement, an image-side surface of the second lens is convex in the paraxial region; and the camera optical lens further satisfies following conditions: $0.50 \leq f_2/f \leq 1.92$; $-0.88 \leq (R_3 + R_4)/(R_3 - R_4) \leq 1.68$; and $0.04 \leq d_3/\text{TTL} \leq 0.19$; where f_2 denotes a focal length of the second lens; R_3 denotes a central curvature radius of an object-side surface of the second lens; R_4 denotes a central curvature radius of an image-side surface of the second lens; and d_3 denotes an on-axis thickness of the second lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(9) As an improvement, the object-side surface of the third lens is convex in a paraxial region; and the camera optical lens further satisfies following conditions: $0.46 \leq f_3/f \leq 1.66$; $-2.21 \leq (R_5 + R_6)/(R_5 - R_6) \leq -0.40$; and $0.03 \leq d_5/\text{TTL} \leq 0.20$; where f_3 denotes a focal length of the third lens; R_5 denotes a central curvature radius of an object-side surface of the third lens; R_6 denotes a central curvature radius of an image-side surface of the third lens; d_5 denotes an on-axis thickness of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(10) As an improvement, an image-side surface of the fourth lens is concave in the paraxial region; and the camera optical lens further satisfies following conditions: $-1.45 \leq f_4/f \leq -0.35$; $0.32 \leq (R_7 + R_8)/(R_7 - R_8) \leq 1.59$; and $0.01 \leq d_7/\text{TTL} \leq 0.17$; where f_4 denotes a focal length of the fourth lens; R_7 denotes a central curvature radius of an object-side surface of the fourth lens; R_8 denotes a central curvature radius of an image-side surface of the fourth lens; and d_7 denotes an on-axis thickness of the fourth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(11) As an improvement, an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of the fifth lens is concave in the paraxial region, and the camera optical lens further satisfies following conditions: $0.59 \leq f_5/f \leq 2.32$; $-9.84 \leq (R_9 + R_{10})/(R_9 - R_{10}) \leq -0.69$; and $0.03 \leq d_9/\text{TTL} \leq 0.15$; where f_5 denotes a focal length of the fifth lens; d_9 denotes an on-axis thickness of the fifth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(12) As an improvement, an object-side surface of the sixth lens is concave in a paraxial region, an image-side surface of the sixth lens is convex in the paraxial region, and the camera optical lens further satisfies following conditions: $-3.82 \leq f_6/f \leq -1.01$; $-5.90 \leq (R_{11} + R_{12})/(R_{11} - R_{12}) \leq -0.71$; and $0.02 \leq d_{11}/\text{TTL} \leq 0.22$; where f_6 denotes a focal length of the sixth lens; R_{11} denotes a central curvature radius of an object-side surface of the sixth lens; R_{12} denotes a central curvature radius of an image-side surface of the sixth lens; d_{11} denotes an on-axis thickness of the sixth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

(13) As an improvement, at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens and the sixth lens is glass.

(14) The present disclosure is advantageous in: the camera optical lens according to the present disclosure has good optical characteristics, has characteristics of large aperture and long focal length, and is especially fit for a mobile phone camera optical lens component and a WEB camera lens composed by such camera elements as a charge coupled device (CCD) and a complementary metal oxide semiconductor (CMOS) for high pixels.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to illustrate the technical solutions in the embodiments of the present disclosure more clearly, the drawings used in the description of the embodiments will be briefly described below. It is obvious that the drawings in the following description are only some embodiments of the present disclosure. For those skilled in the art, other drawings may also be obtained in accordance with the drawings without any inventive effort.

(2) FIG. 1 is a schematic diagram of a structure of a camera optical lens according to Embodiment 1 of the present disclosure.

(3) FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1.

(4) FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1.

(5) FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1.

(6) FIG. 5 is a schematic diagram of a structure of a camera optical lens according to Embodiment 2 of the present disclosure.

(7) FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5.

(8) FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5.

(9) FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5.

(10) FIG. 9 is a schematic diagram of a structure of a camera optical lens according to Embodiment 3 of the present disclosure.

(11) FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9.

(12) FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9.

(13) FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9.

(14) FIG. 13 is a schematic diagram of a structure of a camera optical lens according to a contrasting embodiment of the present disclosure.

(15) FIG. 14 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 13.

(16) FIG. 15 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 13.

(17) FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 13.

DETAILED DESCRIPTION OF EMBODIMENTS

(18) To make the objects, technical solutions, and advantages of the present disclosure clearer, embodiments of the present disclosure are described in detail with reference to accompanying drawings in the following. A person of ordinary skill in the art can understand that, in the embodiments of the present disclosure, many technical details are provided to make readers better understand the present disclosure. However, even without these technical details and any changes and modifications based on the following embodiments, technical solutions required to be protected by the present disclosure can be implemented.

Embodiment 1

(19) Referring to the accompanying drawings, the present disclosure provides a camera optical lens 10. FIG. 1 shows the camera optical lens 10 of Embodiment 1 of the present disclosure, and the camera optical lens 10 includes six lenses. Specifically, the camera optical lens 10 includes, from an object side to an image side: a first lens L1, an aperture S1, a second lens L2, a third lens L3, a fourth lens L4, a fifth lens L5 and a sixth lens L6. An optical element such as an optical filter GF can be arranged between the sixth lens L6 and an image surface Si. In this embodiment, two optical filters are included: GF1 and GF2.

(20) In an embodiment, both an object-side surface and an image-side surface of the first lens L1 are aspheric surfaces. Both an object-side surface and an image-side surface of the sixth lens L6 are aspheric surfaces. Surfaces of the rest lenses are spherical surfaces. By designing some lenses as spherical-surface lenses, difficulty in manufacturing of the lenses may be reduced.

(21) At least one of the first lens L1, the second lens L2, the third lens L3, the fourth lens L4, the fifth lens L5 and the sixth lens L6 is glass. In an embodiment, all of the first lens L1, the second lens L2, the third lens L3, the fourth lens L4, the fifth lens L5 and the sixth lens L6 are glass. Glass lenses facilitate improving performance and stability of the camera optical lens 10. In an alternative embodiment, the lenses may be other materials.

(22) In an embodiment, a focal length of the camera optical lens 10 is defined as f , a focal length of the first lens L1 is defined as f_1 , and the camera optical lens 10 satisfies a condition of $-8.00 \leq f_1/f \leq -3.00$, which specifies a ratio of the focal length f_1 of the first lens L1 and the focal length f of the camera optical lens 10. This range may facilitate effectively balancing a field curvature of the camera optical lens 10, so that a field curvature offset of a central field is less than $20\ \mu\text{m}$.

(23) In an embodiment, a combined focal length of the third lens L3 and the fourth lens L4 is defined as f_{34} , and the camera optical lens 10 satisfies a condition of $-6.00 \leq f_{34}/f \leq -1.50$, which specifies a ratio of the combined focal length f_{34} of the third lens L3 and the fourth lens L4 and the focal length f of the camera optical lens 10. This range may reduce sensitivity of the camera optical lens 10 through a reasonable distribution of focal lengths.

(24) In an embodiment, a central curvature radius of an object-side surface of the fifth lens L5 is defined as R_9 , a central curvature radius of an image-side surface of the fifth lens L5 is defined as R_{10} , and the camera optical lens 10 further satisfies a condition of $R_{10}/R_9 \geq 1.50$, which specifies a shape of the fifth lens L5. The range facilitates correcting astigmatism and distortion of the camera optical lens 10, so that the distortion satisfies a condition of $|\text{Distortion}| \leq 3.0\%$ and possibility of generation of vignetting is reduced.

(25) In an embodiment, a central curvature radius of an object-side surface of the sixth lens L6 is defined as R_{11} , and a central curvature radius of an image-side surface of the sixth lens L6 is defined as R_{12} , and the camera optical lens 10 satisfies a condition of $2.00 \leq R_{12}/R_{11} \leq 30.00$, which specifies a shape of the sixth lens L6. The range reduces deflection of light, and may effectively correct chromatic aberration so that the chromatic aberration satisfies a condition of $|\text{LC}| \leq 3.5\ \mu\text{m}$.

(26) In an embodiment, an on-axis distance from an image-side surface of the second lens L2 to an object-side surface of the third lens L3 is defined as d_4 , an on-axis distance from an image-side surface of the fourth lens L4 to the object-side surface of the fifth lens L5 is defined as d_8 , and the camera optical lens 10 satisfies a condition of $10.00 \leq d_2/d_4 \leq 50.00$, which specifies a ratio of a space between the fourth lens L4 and the fifth lens L5 and a space between the second lens L2 and the third lens L3. This range facilitates shortening a total optical length of the camera optical lens 10 so as to achieve ultra-thinness.

(27) In an embodiment, the object-side surface of the first lens L1 is convex in a paraxial region, and an image-side surface of the first lens L1 is concave in the paraxial region. The first lens L1 has a negative refractive power. In an alternative embodiment, the object-side surface and the image-side surface of the first lens L1 may be set as other distribution conditions of concave and convex surfaces.

(28) A central curvature radius of the object-side surface of the first lens L1 is defined as R_1 , a central curvature radius of the image-side surface of the first lens L1 is defined as R_2 , and the camera optical lens 10 satisfies a condition of $2.11 \leq (R_1 + R_2)/(R_1 - R_2) \leq 7.56$. By

Preferably, by controlling a shape of the first lens L1, the first lens L1 may effectively correct spherical aberration of the system. Preferably, the camera optical lens **10** satisfies a condition of $3.38 \leq (R1+R2)/(R1-R2) \leq 6.05$.

(29) An on-axis thickness of the first lens L1 is defined as d1, a total optical length of the camera optical lens **10** from the object-side surface of the first lens L1 to an image surface of the camera optical lens **10** along an optical axis is defined as TTL, and the camera optical lens **10** satisfies a condition of $0.04 \leq d1/TTL \leq 0.22$. Within the range, miniaturization may be achieved. Preferably, the camera optical lens **10** satisfies a condition of $0.07 \leq d1/TTL \leq 0.17$.

(30) In an embodiment, an object-side surface of the second lens L2 is convex in the paraxial region, an image-side surface of the second lens L2 is convex in the paraxial region, and the second lens L2 has a positive refractive power. In an alternative embodiment, the object-side surface and image-side surface of the second lens L2 may be set as other distribution conditions of concave and convex surfaces.

(31) The focal length of the second lens L2 is defined as f2, and the camera optical lens **10** further satisfies a condition of $0.50 \leq f2/f \leq 1.92$. By controlling a refractive power of the second lens L2 within a reasonable range, correction of the aberration of the optical system may be achieved. Preferably, the camera optical lens **10** further satisfies a condition of $0.81 \leq f2/f \leq 1.53$.

(32) A central curvature radius of the object-side surface of the second lens L2 is defined as R3, a central curvature radius of an image-side surface of the second lens L2 is defined as R4, and the camera optical lens **10** further satisfies a condition of $-0.88 \leq (R3+R4)/(R3-R4) \leq 1.68$, which specifies a shape of the second lens L2. Within this range, a development towards ultra-thin and wide-angle lenses would facilitate correcting a problem of off-axis aberration. Preferably, the camera optical lens **10** further satisfies a condition of $-0.55 \leq (R3+R4)/(R3-R4) \leq 1.35$.

(33) An on-axis thickness of the second lens L2 is defined as d3, and the camera optical lens **10** further satisfies a condition of $0.04 \leq d3/TTL \leq 0.19$. This can facilitate achieving miniaturization. Preferably, the camera optical lens **10** further satisfies a condition of $0.07 \leq d3/TTL \leq 0.15$.

(34) In an embodiment, an object-side surface of the third lens L3 is convex in the paraxial region, an image-side surface of the third lens L3 is convex in the paraxial region, and the third lens L3 has a positive refractive power. In an alternative embodiment, the object-side surface and image-side surface of the third lens L3 may be set as other distribution conditions of concave and convex surfaces.

(35) A focal length of the third lens L3 is defined as f3, and the camera optical lens **10** further satisfies a condition of $0.46 \leq f3/f \leq 1.66$. An appropriate distribution of the refractive power leads to a better imaging quality and a lower sensitivity of the camera optical lens **10**. Preferably, the camera optical lens **10** further satisfies a condition of $0.73 \leq f3/f \leq 1.33$.

(36) A central curvature radius of the object-side surface of the third lens L3 is defined as R5, a central curvature radius of the image-side surface of the third lens L3 is defined as R6, and the camera optical lens **10** further satisfies a condition of $-2.21 \leq (R5+R6)/(R5-R6) \leq -0.40$. This can effectively control a shape of the third lens L3. Within this range, a development towards ultra-thin and wide-angle lenses would facilitate correcting the problem of off-axis aberration. Preferably, the camera optical lens **10** further satisfies a condition of $-1.38 \leq (R5+R6)/(R5-R6) \leq -0.50$.

(37) An on-axis thickness of the third lens L3 is defined as d5, and the camera optical lens **10** further satisfies a condition of $0.03 \leq d5/TTL \leq 0.20$. This range facilitates achieving miniaturization. Preferably, the camera optical lens **10** further satisfies a condition of $0.05 \leq d5/TTL \leq 0.16$.

(38) In an embodiment, an object-side surface of the fourth lens L4 is concave in the paraxial region, an image-side surface of the fourth lens L4 is concave in the paraxial region, and the fourth lens L4 has a negative refractive power. In an alternative embodiment, the object-side surface and image-side surface of the fourth lens L4 may be set as other distribution conditions of concave and convex surfaces. The fourth lens L4 may have a negative refractive power.

(39) A focal length of the fourth lens L4 is defined as f4, and the camera optical lens **10** further satisfies a condition of $-1.45 \leq f4/f \leq -0.35$. The appropriate distribution of refractive power makes the system have better imaging quality and lower sensitivity. Preferably, the camera optical lens **10** further satisfies a condition of $-0.91 \leq f4/f \leq -0.44$.

(40) A central curvature radius of the object-side surface of the fourth lens L4 is defined as R7, a central curvature radius of the image-side surface of the fourth lens L4 is defined as R8, and the camera optical lens **10** further satisfies a condition of $0.32 \leq (R7+R8)/(R7-R8) \leq 1.59$, which specifies a shape of the fourth lens L4. With a development towards ultra-thin and long-focal-length lenses, this range facilitates correcting a problem like an off-axis aberration. Preferably, the camera optical lens **10** further satisfies a condition of $0.51 \leq (R7+R8)/(R7-R8) \leq 1.27$.

(41) An on-axis thickness of the fourth lens L4 is defined as d7, and the camera optical lens **10** further satisfies a condition of $0.01 \leq d7/TTL \leq 0.17$. This can facilitate achieving miniaturization. Preferably, the camera optical lens **10** further satisfies a condition of $0.02 \leq d7/TTL \leq 0.13$.

(42) In an embodiment, an object-side surface of the fifth lens L5 is convex in the paraxial region, an image-side surface of the fifth lens L5 is concave in the paraxial region, and the fifth lens L5 has a positive refractive power. In an alternative embodiment, the object-side surface and image-side surface of the fifth lens L5 may be set as other distribution conditions of concave and convex surfaces.

(43) A focal length of the fifth lens L5 is defined as f5, and the camera optical lens **10** further satisfies a condition of $0.59 \leq f5/f \leq 2.32$. Through reasonable distribution of refractive powers, the system may have better imaging quality and lower sensitivity. Preferably, the camera optical lens **10** further satisfies a condition of $0.95 \leq f5/f \leq 1.86$.

(44) The fifth lens L5 further satisfies a condition of $-9.84 \leq (R9+R10)/(R9-R10) \leq -0.69$, which specifies a shape of the fifth lens L5. Within this range, a development towards ultra-thin and wide-angle lenses facilitates correcting a problem of the off-axis aberration. Preferably, the camera optical lens **10** further satisfies a condition of $-6.15 \leq (R9+R10)/(R9-R10) \leq -0.87$.

(45) An on-axis thickness of the fifth lens L5 is defined as d9, and the camera optical lens **10** further satisfies a condition of $0.03 \leq d9/TTL \leq 0.15$. This range facilitates achieving miniaturization. Preferably, the camera optical lens **10** further satisfies a condition of $0.05 \leq d9/TTL \leq 0.12$.

(46) In an embodiment, an object-side surface of the sixth lens L6 is concave in the paraxial region, an image-side surface of the sixth lens L6 is convex in the paraxial region, and the sixth lens L6 has a negative refractive power. In an alternative embodiment, the object-side surface and image-side surface of the sixth lens L6 may be set as other distribution conditions of concave and convex surfaces.

(47) A focal length of the sixth lens L6 is defined as f6, and the camera optical lens **10** further satisfies a condition of $-3.82 \leq f6/f \leq -1.01$. The appropriate distribution of refractive power makes it possible that the system has the better imaging quality and lower sensitivity. Preferably, the camera optical lens **10** further satisfies a condition of $-2.39 \leq f6/f \leq -1.26$.

(48) The sixth lens L6 further satisfies a condition of $-5.90 \leq (R11+R12)/(R11-R12) \leq -0.71$, which specifies a shape of the sixth lens L6. With a development towards ultra-thin and wide-angle lenses, this range facilitates correcting a problem of off-axis aberration. Preferably,

the camera optical lens **10** further satisfies a condition of $-3.69 \leq (R11+R12)/(R11-R12) \leq -0.89$.

(49) An on-axis thickness of the sixth lens L6 is defined as d11, and the camera optical lens **10** further satisfies a condition of $0.02 \leq d11/TTL \leq 0.22$. This range facilitates achieving miniaturization. Preferably, the camera optical lens **10** further satisfies a condition of $0.03 \leq d11/TTL \leq 0.17$.

(50) In an embodiment, an image height of the camera optical lens **10** is IH, an effective focal length of the camera optical lens **10** is EFL, and the camera optical lens **10** satisfies a condition of $EFL/IH \geq 3.20$. Therefore, the camera optical lens **10** has a long focal length.

(51) In an embodiment, an F number of the camera optical lens **10** is FNO which is less than or equal to 1.65. Thus, the camera optical lens **10** has a large aperture and a better imaging performance. Preferably, the F number of FNO of the camera optical lens **10** is less than or equal to 1.60.

(52) The camera optical lens **10** has outstanding optical functions, while satisfying a desire of large aperture and long focal length.

According to characteristics of the camera optical lens **10**, the camera optical lens **10** is especially fit for a mobile phone camera optical lens component and a WEB camera lens composed by such camera elements as CCD and CMOS for high pixels.

(53) In the following, examples will be used to describe the camera optical lens **10** of the present disclosure. The symbols recorded in each example will be described as follows. The focal length, on-axis distance, central curvature radius, on-axis thickness, inflexion point position, and arrest point position are all in units of mm.

(54) TTL: Total optical length (on-axis distance from the object-side surface of the first lens L1 to the image surface Si) of the camera optical lens **10** in mm.

(55) F number FNO: A ratio of an effective focal length of the camera optical length and an entrance pupil diameter.

(56) Preferably, inflexion points and/or arrest points can be arranged on the object-side surface and/or the image-side surface of the lens, so as to satisfy the demand for high quality imaging. The description below can be referred for specific implementations.

(57) The design data of the camera optical lens **10** in Embodiment 1 of the present disclosure are shown in Table 1 and Table 2.

(58) TABLE-US-00001 TABLE 1 R d nd vd S1 ∞ d0= -11.331 R1 32.916 d1= 4.000 nd1 1.8017 v1 40.63 R2 20.326 d2= 7.381 R3 20.000 d3= 4.000 nd2 1.6204 v2 60.37 R4 -20.000 d4= 0.070 R5 9.966 d5= 3.479 nd3 1.6204 v3 60.37 R6 -63.805 d6= 0.000 R7 -63.805 d7= 3.286 nd4 1.7552 v4 27.55 R8 7.080 d8= 1.157 R9 11.797 d9= 2.060 nd5 1.7995 v5 42.25 R10 28.359 d10= 2.489 R11 -11.067 d11= 1.200 nd6 1.5806 v6 60.08 R12 -61.773 d12= 0.500 R13 ∞ d13= 0.500 ndg1 1.5233 vg1 54.52 R14 ∞ d14= 0.900 R15 ∞ d15= 0.500 ndg2 1.5168 vg2 64.17 R16 ∞ d16= 0.300

(59) In the table, meanings of various symbols will be described as follows. S1: aperture; R: curvature radius at a center an optical surface; R1: central curvature radius of the object-side surface of the first lens L1; R2: central curvature radius of the image-side surface of the first lens L1; R3: central curvature radius of the object-side surface of the second lens L2; R4: central curvature radius of the image-side surface of the second lens L2; R5: central curvature radius of the object-side surface of the third lens L3; R6: central curvature radius of the image-side surface of the third lens L3; R7: central curvature radius of the object-side surface of the fourth lens L4; R8: central curvature radius of the image-side surface of the fourth lens L4; R9: central curvature radius of the object-side surface of the fifth lens L5; R10: central curvature radius of the image-side surface of the fifth lens L5; R11: central curvature radius of the object-side surface of the sixth lens L6; R12: central curvature radius of the image-side surface of the sixth lens L6; R13: central curvature radius of an object-side surface of an optical filter GF1; R14: central curvature radius of an image-side surface of the optical filter GF1; R15: central curvature radius of an object-side surface of an optical filter GF2; R16: central curvature radius of the image-side surface of the optical filter GF2; d: on-axis thickness of a lens and an on-axis distance between lenses; d0: on-axis distance from the aperture S1 to the object-side surface of the first lens L1; d1: on-axis thickness of the first lens L1; d2: on-axis distance from the image-side surface of the first lens L1 to the object-side surface of the second lens L2; d3: on-axis thickness of the second lens L2; d4: on-axis distance from the image-side surface of the second lens L2 to the object-side surface of the third lens L3; d5: on-axis thickness of the third lens L3; d6: on-axis distance from the image-side surface of the third lens L3 to the object-side surface of the fourth lens L4; d7: on-axis thickness of the fourth lens L4; d8: on-axis distance from the image-side surface of the fourth lens L4 to the object-side surface of the fifth lens L5; d9: on-axis thickness of the fifth lens L5; d10: on-axis distance from the image-side surface of the fifth lens L5 to the object-side surface of the sixth lens L6; d11: on-axis thickness of the sixth lens L6; d12: on-axis distance from the image-side surface of the sixth lens L6 to the object-side surface of the optical filter GF1; d13: on-axis thickness of the optical filter GF1; d14: on-axis distance from the image-side surface of the optical filter GF1 to an object-side surface of the optical filter GF2; d15: on-axis thickness of the optical filter GF2; d16: on-axis distance from the image-side surface to the image surface S1 of the optical filter GF2; nd: refractive index of the d line (the d line is a green light having a wavelength of 550 nm); nd1: refractive index of the d line of the first lens L1; nd2: refractive index of the d line of the second lens L2; nd3: refractive index of the d line of the third lens L3; nd4: refractive index of the d line of the fourth lens L4; nd5: refractive index of the d line of the fifth lens L5; nd6: refractive index of the d line of the sixth lens L6; ndg1: refractive index of the d line of the optical filter GF1; ndg2: refractive index of the d line of the optical filter GF2; vd: abbe number; v1: abbe number of the first lens L1; v2: abbe number of the second lens L2; v3: abbe number of the third lens L3; v4: abbe number of the fourth lens L4; v5: abbe number of the fifth lens L5; v6: abbe number of the sixth lens L6; vg1: abbe number of the optical filter GF1; vg2: abbe number of the optical filter GF2.

(60) Table 2 shows aspherical surface data of the camera optical lens **10** in Embodiment 1 of the present disclosure.

(61) TABLE-US-00002 TABLE 2 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 1.5535E+01 -1.5916E-04 -1.1650E-06 7.0513E-08 -3.9764E-09 1.2477E-10 R2 5.5114E+00 -1.4318E-04 -2.5156E-06 3.4268E-07 -2.6001E-08 1.1316E-09 R11 4.3009E+00 -2.8584E-03 9.2300E-05 3.3162E-06 -1.8944E-06 3.1786E-07 R12 1.4051E+02 -3.1288E-03 6.3162E-05 1.3332E-05 -3.4145E-06 4.2900E-07 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 R1 1.5535E+01 -2.3456E-12 2.3156E-14 -9.4202E-17 0.0000E+00 R2 5.5114E+00 -2.7220E-11 2.7022E-13 1.1422E-15 -3.1389E-17 R11 4.3009E+00 -3.0783E-08 1.7802E-09 -5.6882E-11 7.8442E-13 R12 1.4051E+02 -3.2315E-08 1.4577E-09 -3.6217E-11 3.8169E-13

(62) For convenience, an aspheric surface of each lens surface is an aspheric surface shown in the below formula (1). However, the present disclosure is not limited to the aspherical polynomials form shown in the formula (1).

$$z = (cr.\sup.2)/\{1+[1-(k+1)$$

$$(c.\sup.2r.\sup.2)].\sup.1/2\} + A4r.\sup.4 + A6r.\sup.6 + A8r.\sup.8 + A10r.\sup.10 + A12r.\sup.12 + A14r.\sup.14 + A16r.\sup.16 + A18r.\sup.18 + A20r.\sup.20$$

(1)

(63) Herein, k is a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18 and A20 are aspheric surface coefficients, c is a curvature at a center of the optical surface, r is a vertical distance from a point on an aspheric surface curve to the optical axis, and z is an aspheric

surface depth (a vertical distance between a point on the aspheric surface which is of the distance of r from the optical axis, and a tangent surface that is tangent with a top point of the optical axis of the aspheric surface).

(64) Table 3 shows design data of inflexion points of the camera optical lens **10** according to Embodiment 1 of the present disclosure. Herein, P1R1 represents the object-side surface of the first lens L1. The data in the column named “inflexion point position” refer to vertical distances from inflexion points arranged on each lens surface to the optic axis of the camera optical lens **10**.

(65) TABLE-US-00003 TABLE 3 Inflexion point number Inflexion point position 1 P1R1 1 5.095

(66) FIG. 2 and FIG. 3 illustrate a longitudinal aberration and a lateral color with wavelengths of 650 nm, 610 nm, 550 nm, 510 nm, 470 nm and 430 nm after passing the camera optical lens **10** according to Embodiment 1, respectively. FIG. 4 illustrates a field curvature and a distortion with a wavelength of 550 nm after passing the camera optical lens **10** according to Embodiment 1. A field curvature S in FIG. 4 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

(67) Table 16 shows values in the embodiments corresponding to the parameters specified in the conditions.

(68) As shown in Table 16, Embodiment 1 satisfies the conditions.

(69) In an embodiment, an entrance pupil diameter ENPD of the camera optical lens **10** is 9.611 mm, an image height IH of 1.0H is 4.626 mm, and an FOV (field of view) in a diagonal direction is 34.38°. Thus, the camera optical lens **10** satisfies a desire of design in large aperture and long focal length. Its on-axis and off-axis aberrations are fully corrected, thereby achieving excellent optical characteristics.

Embodiment 2

(70) Embodiment 2 is basically the same as Embodiment 1 and involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

(71) In an embodiment, the object-side surface of the second lens L2 is concave in the paraxial region.

(72) FIG. 5 is a camera optical lens **20** according to Embodiment 2 of the present disclosure.

(73) Table 4 and Table 5 show design data of a camera optical lens **20** in Embodiment 2 of the present disclosure.

(74) TABLE-US-00004 TABLE 4 R d nd vd S1 ∞ d0= -9.076 R1 17.051 d1= 2.465 nd1 1.8017 v1 40.63 R2 10.937 d2= 6.744 R3 -200.000 d3= 2.465 nd2 1.6204 v2 60.37 R4 -11.417 d4= 0.029 R5 11.552 d5= 4.000 nd3 1.6204 v3 60.37 R6 -45.674 d6= 0.002 R7 -45.674 d7= 0.841 nd4 1.7552 v4 27.55 R8 10.296 d8= 0.579 R9 15.083 d9= 1.990 nd5 1.7995 v5 42.25 R10 800.000 d10= 7.420 R11 -7.500 d11= 0.992 nd6 1.5806 v6 60.08 R12 -15.200 d12= 0.600 R13 ∞ d13= 0.500 ndg1 1.5233 vg1 54.52 R14 ∞ d14= 0.800 R15 ∞ d15= 0.500 ndg2 1.5168 vg2 64.17 R16 ∞ d16= 0.104

(75) Table 5 shows aspherical surface data of each lens of the camera optical lens **20** in Embodiment 2 of the present disclosure.

(76) TABLE-US-00005 TABLE 5 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -4.0893E+00 -2.3688E-04 -4.2406E-06 4.6263E-08 -3.5705E-09 1.8138E-10 R2 2.2397E+00 -5.3230E-04 -5.0214E-06 -7.7150E-07 1.1344E-07 -1.0571E-08 R11 -8.7189E+00 3.1136E-03 -2.9717E-04 3.5119E-05 -3.3458E-06 2.1717E-07 R12 -1.9900E+02 6.7294E-04 7.4657E-04 -1.7164E-04 2.4976E-05 -2.4096E-06 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 R1 -4.0893E+00 -4.9541E-12 6.2628E-14 -2.5233E-16 0.0000E+00 R2 2.2397E+00 5.9550E-10 -2.0308E-11 3.8241E-13 -3.0984E-15 R11 -8.7189E+00 -9.2722E-09 2.5189E-10 -3.9478E-12 2.6939E-14 R12 -1.9900E+02 1.5259E-07 -6.0625E-09 1.3642E-10 -1.3144E-12

(77) Table 6 and Table 7 show design data of inflexion points and arrest points of lenses in the camera optical lens **10** according to Embodiment 2 of the present disclosure. Herein, P1R1 and P1R2 respectively represent the object-side surface and the image-side surface of the first lens L1, P2R1 and P2R2 respectively represent the object-side surface and the image-side surface of the second lens L2, P3R1 and P3R2 respectively represent the object-side surface and the image-side surface of the third lens L3, P4R1 and P4R2 respectively represent the object-side surface and the image-side surface of the fourth lens L4, P5R1 and P5R2 respectively represent the object-side surface and the image-side surface of the fifth lens L5, and P6R1 and P6R2 respectively represent the object-side surface and the image-side surface of the sixth lens L6. The data in the column named “inflexion point position” refer to vertical distances from inflexion points arranged on each lens surface to the optic axis of the camera optical lens **20**. The data in the column named “arrest point position” refer to vertical distances from arrest points arranged on each lens surface to the optical axis of the camera optical lens **20**.

(78) TABLE-US-00006 TABLE 6 Inflexion point number Inflexion point position 1 P1R1 1 3.425 P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 1 1.885 P6R2 1 1.035

(79) TABLE-US-00007 TABLE 7 Arrest point number Arrest point position 1 P1R1 1 5.555 P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 1 4.185 P6R2 1 1.905

(80) FIG. 6 and FIG. 7 illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 610 nm, 550 nm, 510 nm, 470 nm and 430 nm after passing the camera optical lens **20** according to Embodiment 2. FIG. 8 illustrates a field curvature and a distortion of light with a wavelength of 550 nm after passing the camera optical lens **20** according to Embodiment 2. A field curvature S in FIG. 8 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

(81) As shown in Table 16, Embodiment 2 satisfies the conditions.

(82) In an embodiment, an entrance pupil diameter ENPD of the camera optical lens **20** is 9.455 mm, an image height IH of 1.0H is 4.626 mm, and an FOV (field of view) in the diagonal direction is 33.81°. Thus, the camera optical lens **20** satisfies a desire of design in large aperture and long focal length. Its on-axis and off-axis aberrations are fully corrected, thereby achieving excellent optical characteristics.

Embodiment 3

(83) Embodiment 3 is basically the same as Embodiment 1 and involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

(84) In an embodiment, the image-side surface of the third lens L3 is concave in the paraxial region and the object-side surface of the fourth lens L4 is convex in the paraxial region.

(85) FIG. 9 is a camera optical lens **30** according to Embodiment 3 of the present disclosure.

(86) Table 8 and Table 9 show design data of the camera optical lens **30** in Embodiment 3 of the present disclosure.

(87) TABLE-US-00008 TABLE 8 R d nd vd S1 ∞ d0= -10.611 R1 41.985 d1= 4.000 nd1 1.8017 v1 40.63 R2 28.089 d2= 5.621 R3 12.817 d3= 2.282 nd2 1.6204 v2 60.37 R4 -32.853 d4= 0.050 R5 9.888 d5= 1.811 nd3 1.6204 v3 60.37 R6 200.000 d6= 0.002 R7 200.000 d7= 3.103 nd4 1.7552 v4 27.55 R8 5.776 d8= 0.251 R9 6.550 d9= 2.684 nd5 1.7995 v5 42.25 R10 9.891 d10= 1.767 R11 -16.098 d11= 4.000 nd6 1.5806 v6 60.08 R12 -480.000 d12= 0.200 R13 ∞ d13= 0.400 ndg1 1.5233 vg1 54.52 R14 ∞ d14= 1.000 R15 ∞ d15= 0.400 ndg2 1.5168 vg2 64.17 R16 ∞ d16= 0.129

(88) Table 9 shows aspherical surface data of each lens of the camera optical lens **30** in Embodiment 3 of the present disclosure.

(89) TABLE-US-00009 TABLE 9 Conic coefficient Aspherical surface coefficients k A4 A6 A8 A10 A12 R1 8.2978E+00 -2.1422E-04

-2.7935E-07 2.1609E-08 -5.8891E-10 1.4327E-11 R2 3.5679E+00 -2.2265E-04 1.0342E-06 4.0963E-09 8.8149E-10
-4.9414E-11 R11 6.0394E+00 -2.1460E-03 3.8383E-05 -2.1124E-05 6.2143E-06 -1.3309E-06 R12 4.2268E+01 -1.4614E-03
-6.0096E-05 3.2606E-05 -7.0854E-06 9.3256E-07 Conic coefficient Aspherical surface coefficients k A14 A16 A18 A20 R1
8.2978E+00 -1.4697E-13 5.8811E-16 -1.9919E-18 0.0000E+00 R2 3.5679E+00 1.4867E-12 -2.0991E-14 9.7920E-17
1.7569E-19 R11 6.0394E+00 1.8574E-07 -1.5374E-08 6.8542E-10 -1.2800E-11 R12 4.2268E+01 -7.4808E-08 3.5748E-09
-9.2939E-11 1.0034E-12

(90) Table 10 and Table 11 show design data of inflexion points and arrest points of lenses in the camera optical lens **30** in Embodiment 3 of the present disclosure.

(91) TABLE-US-00010 TABLE 10 Inflexion point number Inflexion point position 1 P1R1 1 3.175

(92) TABLE-US-00011 TABLE 11 Arrest point number Arrest point position 1 P1R1 1 5.705

(93) FIG. **10** and FIG. **11** illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 610 nm, 550 nm, 510 nm, 470 nm and 430 nm after passing the camera optical lens **30** according to Embodiment 3. FIG. **12** illustrates a field curvature and a distortion of light with a wavelength of 550 nm after passing the camera optical lens **30** according to Embodiment 3. A field curvature S in FIG. **12** is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

(94) As shown in Table 16, Embodiment 3 satisfies the conditions.

(95) In an embodiment, an entrance pupil diameter ENPD of the camera optical lens **30** is 9.378 mm, an image height IH of 1.0H is 4.626 mm, and an FOV (field of view) in the diagonal direction is 33.88°. Thus, the camera optical lens **30** satisfies a desire of design in large aperture and long focal length. Its on-axis and off-axis aberrations are fully corrected, thereby achieving excellent optical characteristics.

Contrasting Embodiment

(96) The contrasting embodiment involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

(97) In the contrasting embodiment, the object-side surface of the second lens L2 is concave in the paraxial region.

(98) FIG. **13** is a camera optical lens **40** according to the contrasting embodiment.

(99) Table 12 and Table 13 show design data of the camera optical lens **40** in the contrasting embodiment of the present disclosure.

(100) TABLE-US-00012 TABLE 12 R d nd vd S1 ∞ d0= -7.808 R1 16.732 d1= 1.808 nd1 1.8017 v1 40.63 R2 8.802 d2= 6.319 R3 -200.000 d3= 4.000 nd2 1.6204 v2 60.37 R4 -9.959 d4= 0.029 R5 12.361 d5= 4.000 nd3 1.6204 v3 60.37 R6 -28.100 d6= 0.002 R7 -28.100 d7= 1.035 nd4 1.7552 v4 27.55 R8 11.037 d8= 0.579 R9 15.083 d9= 2.226 nd5 1.7995 v5 42.25 R10 800.000 d10= 5.842 R11 -7.500 d11= 0.800 nd6 1.5806 v6 60.08 R12 -15.200 d12= 1.000 R13 ∞ d13= 0.500 ndg1 1.5233 vg1 54.52 R14 ∞ d14= 0.500 R15 ∞ d15= 0.500 ndg2 1.5168 vg2 64.17 R16 ∞ d16= 3.200

(101) Table 13 shows aspherical surface data of each lens of the camera optical lens **40** in the contrasting embodiment of the present disclosure.

(102) TABLE-US-00013 TABLE 13 Conic coefficient Aspherical surface coefficients k A4 A6 A8 A10 A12 R1 -1.1110E+00 3.9801E-05 -4.6924E-05 5.9431E-06 -4.5826E-07 2.1023E-08 R2 1.7951E+00 -3.2315E-04 4.9331E-05 -2.3750E-05 4.5967E-06 -5.0858E-07 R11 -7.4254E+00 3.8342E-03 -3.4029E-04 4.0528E-05 -4.3057E-06 2.9317E-07 R12 -8.8936E+01 9.4990E-04 8.7441E-04 -2.0368E-04 2.7235E-05 -2.3113E-06 Conic coefficient Aspherical surface coefficients k A14 A16 A18 A20 R1 -1.1110E+00 -5.6794E-10 8.3074E-12 -5.0539E-14 0.0000E+00 R2 1.7951E+00 3.3591E-08 -1.3189E-09 2.8443E-11 -2.6070E-13 R11 -7.4254E+00 -1.1997E-08 2.8268E-10 -3.5164E-12 1.7862E-14 R12 -8.8936E+01 1.2340E-07 -3.9695E-09 6.9562E-11 -5.0667E-13

(103) Table 14 and Table 15 show design data of inflexion points and arrest points of lenses in the camera optical lens **40** in the contrasting embodiment of the present disclosure.

(104) TABLE-US-00014 TABLE 14 Inflexion Inflexion Inflexion point number point position 1 point position 2 P1R1 1 4.275 / P1R2 0 / / P2R1 0 / / P2R2 0 / / P3R1 0 / / P3R2 0 / / P4R1 0 / / P4R2 0 / / P5R1 0 / / P5R2 0 / / P6R1 2 1.755 3.985 P6R2 2 1.115 4.265

(105) TABLE-US-00015 TABLE 15 Arrest point number Arrest point position 1 P6R2 1 1.985

(106) FIG. **14** and FIG. **15** illustrate a longitudinal aberration and a lateral color of light with wavelengths of 650 nm, 610 nm, 550 nm, 510 nm, 470 nm and 430 nm after passing the camera optical lens **40** according to the contrasting embodiment. FIG. **16** illustrates a field curvature and a distortion of light with a wavelength of 550 nm after passing the camera optical lens **40** according to the contrasting embodiment. A field curvature S in FIG. **16** is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

(107) Table 16 in the following lists values corresponding to the respective conditions in the contrasting embodiment according to the above conditions. Obviously, the camera optical lens **40** in the contrasting embodiment does not satisfy the above condition of $-8.00 \leq f1/f \leq -3.00$ and can not effectively balance the field curvature of the camera optical lens **40**.

(108) In the contrasting embodiment, an entrance pupil diameter ENPD of the camera optical lens **40** is 9.514 mm, an image height IH of 1.0H is 4.626 mm, and an FOV (field of view) in the diagonal direction is 33.93°. Thus, the camera optical lens **40** does not satisfy a desire of design in large aperture and long focal length.

(109) TABLE-US-00016 TABLE 16 Parameters and Embodiment Embodiment Embodiment Contrasting conditions 1 2 3 Embodiment f1/f -4.996 -3.051 -7.987 -1.685 f34/f -3.197 -5.955 -1.511 -5.275 R10/R9 2.404 53.040 1.510 53.040 R12/R11 5.582 2.027 29.817 2.027 d8/d4 16.529 19.966 5.020 19.966 f 15.378 15.128 15.004 15.223 f1 -76.834 -46.154 -119.838 -25.657 f2 16.702 19.351 15.098 16.698 f3 14.099 15.217 16.647 14.332 f4 -8.210 -10.970 -7.870 -10.296 f5 23.818 19.110 17.773 19.108 f6 -23.342 -26.675 -28.678 -26.421 f34 -49.170 -90.093 -22.674 -80.307 FNO 1.600 1.600 1.600 1.600 TTL 31.822 30.031 27.700 32.340

(110) It can be appreciated by one having ordinary skill in the art that the description above is only embodiments of the present disclosure. In practice, one having ordinary skill in the art can make various modifications to these embodiments in forms and details without departing from the scope of the present disclosure.

Claims

1. A camera optical lens comprising, from an object side to an image side: a first lens having a negative refractive power; a second lens having a positive refractive power; a third lens having a positive refractive power; a fourth lens having a negative refractive power; a fifth lens having a positive refractive power; and a sixth lens having a negative refractive power; wherein the camera optical lens satisfies following conditions:
 $-8.00 \leq f1/f \leq -3.00$;

$-6.00 \leq f_{34}/f \leq -1.50$; and $R_{10}/R_9 \geq 1.50$; where f denotes a focal length of the camera optical lens; f_1 denotes a focal length of the first lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; R_9 denotes a central curvature radius of an object-side surface of the fifth lens; and R_{10} denotes a central curvature radius of an image-side surface of the fifth lens.

2. The camera optical lens according to claim 1 further satisfying following condition:
 $2.00 \leq R_{12}/R_{11} \leq 30.00$; where R_{11} denotes a central curvature radius of an object-side surface of the sixth lens; and R_{12} denotes a central curvature radius of an image-side surface of the sixth lens.

3. The camera optical lens according to claim 1 further satisfying following condition:
 $5.00 \leq d_8/d_4 \leq 20.00$; where d_4 denotes an on-axis distance from an image-side surface of the second lens to an object-side surface of the third lens; and d_8 denotes an on-axis distance from an image-side surface of the fourth lens to an object-side surface of the fifth lens.

4. The camera optical lens according to claim 1, wherein an object-side surface of the first lens is convex in a paraxial region and an image-side surface of the first lens is concave in the paraxial region; and the camera optical lens further satisfies following conditions:
 $2.11 \leq (R_1 + R_2)/(R_1 - R_2) \leq 7.56$; and
 $0.04 \leq d_1/\text{TTL} \leq 0.22$; where R_1 denotes a central curvature radius of an object-side surface of the first lens; R_2 denotes a central curvature radius of an image-side surface of the first lens; d_1 denotes an on-axis thickness of the first lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

5. The camera optical lens according to claim 1, wherein an image-side surface of the second lens is convex in the paraxial region; and the camera optical lens further satisfies following conditions:
 $0.50 \leq f_2/f \leq 1.92$;
 $-0.88 \leq (R_3 + R_4)/(R_3 - R_4) \leq 1.68$; and
 $0.04 \leq d_3/\text{TTL} \leq 0.19$; where f_2 denotes a focal length of the second lens; R_3 denotes a central curvature radius of an object-side surface of the second lens; R_4 denotes a central curvature radius of an image-side surface of the second lens; and d_3 denotes an on-axis thickness of the second lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

6. The camera optical lens according to claim 1, wherein the object-side surface of the third lens is convex in a paraxial region; and the camera optical lens further satisfies following conditions:
 $0.46 \leq f_3/f \leq 1.66$;
 $-2.21 \leq (R_5 + R_6)/(R_5 - R_6) \leq -0.40$; and
 $0.03 \leq d_5/\text{TTL} \leq 0.20$; where f_3 denotes a focal length of the third lens; R_5 denotes a central curvature radius of an object-side surface of the third lens; R_6 denotes a central curvature radius of an image-side surface of the third lens; d_5 denotes an on-axis thickness of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

7. The camera optical lens according to claim 1, wherein an image-side surface of the fourth lens is concave in the paraxial region; and the camera optical lens further satisfies following conditions:
 $-1.45 \leq f_4/f \leq -0.35$;
 $0.32 \leq (R_7 + R_8)/(R_7 - R_8) \leq 1.59$; and
 $0.01 \leq d_7/\text{TTL} \leq 0.17$; where f_4 denotes a focal length of the fourth lens; R_7 denotes a central curvature radius of an object-side surface of the fourth lens; R_8 denotes a central curvature radius of an image-side surface of the fourth lens; and d_7 denotes an on-axis thickness of the fourth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

8. The camera optical lens according to claim 1, wherein an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of the fifth lens is concave in the paraxial region, and the camera optical lens further satisfies following conditions:
 $0.59 \leq f_5/f \leq 2.32$;
 $-9.84 \leq (R_9 + R_{10})/(R_9 - R_{10}) \leq -0.69$; and
 $0.03 \leq d_9/\text{TTL} \leq 0.15$; where f_5 denotes a focal length of the fifth lens; d_9 denotes an on-axis thickness of the fifth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

9. The camera optical lens according to claim 1, wherein an object-side surface of the sixth lens is concave in a paraxial region, an image-side surface of the sixth lens is convex in the paraxial region, and the camera optical lens further satisfies following conditions:
 $-3.82 \leq f_6/f \leq -1.01$;
 $-5.90 \leq (R_{11} + R_{12})/(R_{11} - R_{12}) \leq -0.71$; and
 $0.02 \leq d_{11}/\text{TTL} \leq 0.22$; where f_6 denotes a focal length of the sixth lens; R_{11} denotes a central curvature radius of an object-side surface of the sixth lens; R_{12} denotes a central curvature radius of an image-side surface of the sixth lens; d_{11} denotes an on-axis thickness of the sixth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image surface of the camera optical lens along an optical axis.

10. The camera optical lens according to claim 1, wherein at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens and the sixth lens is glass.
